

Mahabharata Explorer

Phase 2 — Consolidation & Architecture Audit

Complete 133-Point Audit

PHASE 2 — CONSOLIDATION & ARCHITECTURE AUDIT

The purpose of this phase

We're going to answer six questions:

1. What have we actually decided?

2. Which things are shared systems rather than individual features?
3. Where are we accidentally designing the same thing twice?
4. Are any decisions conflicting?
5. What belongs in V1, V2, and later?
6. What exact architecture should Base44 be instructed to build?

The goal is to emerge with:

One coherent Master Blueprint.

1. THE BIGGEST ARCHITECTURAL DISCOVERY

Looking back at everything we've designed, I think the project is actually composed of five layers, not simply a collection of pages.

Data

Evidence

Presentation

Exploration

User State

2. LAYER 1 — KNOWLEDGE / DATA

This is the actual world we're representing.

Core entities:

Characters

Events

Locations

Groups

Relationships

Family Relationships

Wars

War Days

Formations

Sources

Claims

Evidence

The most important conclusion:

These are entities in one connected knowledge graph.

3. LAYER 2 — EVIDENCE

This is what we established in Phase 2.11.

Claim → Evidence → Source

Evidence isn't a separate "Sources page."

It's a cross-cutting layer underneath the entire product.

4. LAYER 3 — EXPLORATION

This is how users navigate the knowledge graph.

Timeline

Character Explorer

Event Explorer

Relationship Explorer

Family Tree

Map

War Explorer

Search

These should not each invent independent interaction systems.

5. LAYER 4 — PRESENTATION

This is the visual layer:

Typography

Colors

Cards

Panels

Diagrams

Animation

Responsive layouts

Mahabharata-inspired assets

And this is where our "Modern first, traditional second" principle lives.

6. LAYER 5 — USER STATE

Examples:

Focus

Selected Day

Selected Event

Active Filters

Search Query

Theme

Density

Text Size

Motion

Visual Character

This distinction is extremely important.

7. THE MOST IMPORTANT CONSOLIDATION

I would now formally define:

Data = What exists.

Evidence = Why it is represented.

Exploration = How the user navigates it.

Presentation = How it looks.

State = What the user is currently doing/prefering.

8. THE MASTER KNOWLEDGE GRAPH

The application should fundamentally behave as one connected knowledge graph:

Source

↓

Evidence

↓

Claim

↓

Entity

Entities connect as:

Character

Event

Location

Group

Relationship

War

War Day

Formation

Those are exposed through:

Timeline

Map

Explorers

Focus

Search

9. DUPLICATION AUDIT

This is where I see some potentially dangerous duplication.

Character Journey and War Journey should not be two systems.

Character Journey = general journey.

War Journey = filtered/contextual Character Journey.

10. WAR EVENTS

We have Events and War Events.

There should be only one Event entity.

A War Event is simply:

Event + war context + war day context, where applicable.

11. MAPS

Main Map
War Map
Character Journey Map
Location Map

These shouldn't be separate map engines.

They should be one Map system with different datasets/layers/context.

12. TIMELINES

Main Timeline
War Timeline
Character Journey Timeline
Day Timeline

These should share one timeline visualization architecture.

13. SOURCES

Event sources
Character sources
War sources
Formation sources
Map sources

These are not separate source systems.

They all connect to Source → Evidence → Claim.

14. FOCUS

Character Focus
Event Focus
War Focus
Map Focus

These aren't different features.

They are one global Focus system operating on different entity types.

15. SEARCH

We don't want independent Character Search, Event Search, War Search, Source Search, and Location Search.

Instead:

Global Search with entity type, context, ranking, and filters.

16. MASTER REUSABLE SYSTEMS

1. Navigation System

2. Timeline System
3. Map System
4. Entity System
5. Relationship System
6. Event System
7. Search System
8. Focus System

- 9. Evidence System
- 10. Source System
- 11. Filtering System
- 12. Settings/Preferences System
- 13. Responsive Layout System
- 14. Design System

17. DESIGN SYSTEM AUDIT

We need to prevent every phase from inventing its own card, button, panel, modal, drawer, badge, filter, timeline marker, etc.

We need a shared component vocabulary.

18. CORE UI COMPONENTS

Navigation:

- Global Header
- Desktop Navigation
- Mobile Navigation
- Breadcrumbs
- Context Navigation

Content:

- Entity Card
- Event Card
- Character Card
- Location Card
- Source Card
- Summary Card

Interaction:

- Button
- Icon Button
- Segmented Control
- Tabs
- Filter
- Search
- Dropdown
- Toggle

Overlays:

- Drawer
- Bottom Sheet
- Modal
- Tooltip
- Popover
- Toast

Data visualization:

- Timeline
- Timeline Marker
- Relationship Node
- Relationship Edge

Map Marker
Journey Path
Formation Diagram

States:
Loading
Empty
Error
No Results
Partial Data

19. THE CARD PROBLEM

Avoid card explosion.

Use a Base Entity Card with variants:

compact
standard
featured
horizontal

and flexible content slots.

20. DRAWERS

Don't build separate Character Drawer, Event Drawer, Source Drawer, Location Drawer, War Drawer.

Build a reusable Context Drawer, with a Context Bottom Sheet on mobile.

21. DETAIL PAGES

Use a shared Entity Detail Layout.

Potential sections:

Entity Header
Summary
Key Information
Relationships
Timeline
Map
Evidence
Related Items

Not every entity needs every section.

22. INFORMATION ARCHITECTURE AUDIT

The overall information architecture:

Home / Entry
Timeline
Explore
Characters
Events
Locations
Relationships

Family Tree
Map
War Explorer
Search

23. ENTITY DETAIL

Every entity detail can connect to:

Related Events
Related People
Related Places
Timeline
Map
Evidence

24. USER-FACING NAVIGATION

Do not expose our internal database architecture as the primary navigation.

The user should not have to choose between:

Sources
Claims
Evidence
Relationships
War Days
etc.

Navigation should be organized around exploration tasks.

25. MAIN NAVIGATION

A cleaner conceptual navigation is:

Explore
Timeline
Map
War
Search

Characters, Events, Relationships, and Sources are discoverable within the explorers.

26. DESKTOP NAVIGATION

Use a persistent navigation rail where information density justifies it, but don't force it onto every screen.

27. MOBILE NAVIGATION

Do not reproduce the desktop rail on mobile.

Use compact top navigation plus contextual bottom/action navigation where useful.

28. RESPONSIVE AUDIT

Responsive is a hard project requirement.

The application must work beautifully in:

Desktop landscape

Tablet landscape
Tablet portrait
Mobile portrait
Mobile landscape

Responsive means the interaction model adapts, not merely that columns become narrower.

29. DESKTOP VS MOBILE

Examples:

Desktop side panel → Mobile bottom sheet.
Desktop horizontal timeline → Mobile-adapted timeline.
Desktop persistent filters → Mobile filter button + sheet.
Desktop rail → Mobile navigation.

30. VISUAL SYSTEM AUDIT

Locked visual principles:

Modern first.
Traditional second.
Flat / dimensional hybrid.
No excessive parchment.
No brown-heavy palette.
No generic "ancient India" aesthetic.
No excessive gold.
No temple/fantasy-game UI.
Indic details as accents.

31. MAHABHARATA ASSET AUDIT

The asset strategy should be:

Generic Component

↓

Theme/Asset Slot

↓

Mahabharata Asset

This allows development to start before the asset hunt is complete.

32. ASSET SLOTS

An Asset Slot can conceptually contain:

component
purpose
default asset
thematic asset
fallback

Example:

Loading Indicator → Generic spinner → Sudarshan Chakra → Static reduced-motion fallback.

33. EVIDENCE AUDIT

Phase 2.11 is a core V1 architecture even if some evidence UI is initially minimal.

34. SOURCE EXPANSION

The original PDF is not the complete source universe.

Initial Dataset + Additional Credible Sources + Future Source Expansion

→ Unified Knowledge Base

35. CREDIBILITY RULE

New information should enter the knowledge base only through identifiable, credible sources, with provenance retained.

The workflow is:

Research → verify → source → structure → evidence → database.

36. BASE44 SHOULD NOT INVENT THE KNOWLEDGE BASE

The eventual Base44 prompt should explicitly say:

Build the application architecture and data model to consume the supplied structured dataset. Do not fabricate missing Mahabharata information.

37. AI-GENERATED CONTENT

Base44 should not fill empty fields with plausible-looking content.

If data is missing:

show an appropriate empty/unknown state.

38. DATA COMPLETENESS

Distinguish:

Known

Unknown

Not yet researched

Not applicable

Conflicting

Approximate

A blank field should never automatically mean "unknown."

39. EDGE STATE AUDIT

Every major component needs:

Loading

Empty

Error

Partial data

No search results

Unavailable source

Unsupported visualization

Mobile fallback

40. MAP EDGE CASE

If a location has no coordinates:
do not crash and do not invent coordinates.

Show that location information exists but map placement is unavailable.

41. TIMELINE EDGE CASE

If an event has no exact date:
do not fabricate a timestamp.

Use relative sequence or approximate position.

42. SOURCE EDGE CASE

If a source has no online URL:
show that a source reference is available rather than a broken Open Source button.

43. CHARACTER EDGE CASE

If no portrait exists:
use an elegant neutral placeholder, not an AI-generated face.

44. FORMATION EDGE CASE

If textual information exists but reliable visualization is not possible:
show the textual description and indicate that visualization is unavailable.

45. PERFORMANCE AUDIT

The project can become heavy because of:
large event sets
relationships
maps
images
timelines
diagrams
sources
animations

We should not load everything at once.

46. PROGRESSIVE LOADING

Load:
Application shell
↓
Primary content
↓
Secondary relationships
↓
Heavy visualizations

47. LAZY LOADING

Especially lazy-load when appropriate:
images
maps

complex diagrams
large relationship graphs

48. TIMELINE PERFORMANCE

Use level-of-detail rendering.

At broad zoom:
aggregated markers.

At close zoom:
individual events.

49. WAR PERFORMANCE

Do not load every day's full event structures if the user is viewing one day.

Load relevant detail progressively.

50. RELATIONSHIP GRAPH PERFORMANCE

At overview:
show major relationships / limited nodes.

On zoom or focus:
expand.

51. MAP PERFORMANCE

Do not render every possible marker at once.

Use layers, clustering, and contextual filtering.

52. ACCESSIBILITY AUDIT

Accessibility is foundational, not a later QA pass.

The application must support:

keyboard navigation
screen readers
visible focus
semantic controls
contrast
reduced motion
responsive text
non-color-only meaning

53. DESIGN TOKENS

Establish a central Design Token System containing:

Colors
Typography
Spacing
Radius
Shadows
Borders
Motion

Breakpoints
Z-index

54. TOKENIZED COMPONENTS

Every component should use shared tokens instead of hardcoded one-off values.

55. COLOR SYSTEM

Use semantic tokens:

background
surface
surface-elevated
text-primary
text-secondary
border
accent
success
warning
error
info

56. MAHABHARATA PALETTE

The thematic palette should also be tokenized.

Theme Layer



Semantic Color Tokens



Components

57. TYPOGRAPHY SYSTEM

Centralize:

display
heading
subheading
body
caption
metadata
labels

58. SPACING SYSTEM

Use a consistent spacing scale.

59. RADIUS SYSTEM

Use a consistent radius system so the interface feels like one product.

60. MOTION TOKENS

Define a small set of motion tokens such as:

fast

standard
slow
reduced-motion

61. RESPONSIVE BREAKPOINT SYSTEM

Use centralized breakpoints.

Prefer content-driven breakpoints over rigid device labels where practical.

62. NAVIGATION AUDIT

Every screen should answer:

Where did I come from?

Where am I?

Where can I go?

How do I return?

63. BREADCRUMBS

Deep contexts should support breadcrumbs, for example:

Mahabharata → War → Day 13 → Event

On mobile this can collapse to a contextual back label.

64. FOCUS AUDIT

Focus is powerful but can become confusing if multiple states are active.

Example:

Focus: Arjuna

Filter: Day 13

Search: Karna

Selected Event: X

The UI needs a clear active-state hierarchy.

65. STATE HIERARCHY

Global:

Focus

Page:

Selected tab / day / view

Local:

Selected event / node / marker

Temporary:

Search / filter / drawer

66. STATE PERSISTENCE RULE

Selecting a day should not silently clear a character focus unless explicitly intended.

Opening a source drawer should not clear the selected event.

67. URL / STATE SYNCHRONIZATION

Major navigational states should be representable in URLs where practical.

Examples:

character/arjuna

war/kurukshetra/day/13

This supports sharing, bookmarks, browser history, and reload persistence.

68. NOT EVERY STATE BELONGS IN URL

Do not encode transient states such as hover or purely visual drawer state unless there is a strong product reason.

69. SHAREABILITY

Users should eventually be able to share:

a character

an event

a specific day

a location

a war view

70. SEARCH → DEEP LINK

Search result for Arjuna should open the character's canonical detail route, not merely a temporary modal.

71. SOURCE → DEEP LINK

Source detail should eventually be directly addressable.

72. RESPONSIVE DEEP LINKS

The same canonical URL should work across desktop and mobile.

73. SECURITY / DATA INTEGRITY

Core rules:

Frontend must not be the source of truth.

Validation belongs at the data layer.

User-editable settings must be validated.

Source metadata must not be silently modified.

Generated/AI content must be distinguished from source-derived content.

74. USER EDITING

V1 recommendation:

No public editing.

The knowledge base is curated.

75. WHY NO PUBLIC EDITING

Public editing would require:

moderation

revision history

contributor accounts

conflict resolution

vandalism prevention

source validation

That is effectively a different product.

76. ADMIN / EDITOR SYSTEM

Eventually build a private editorial layer for:

add source

add character

edit event

attach evidence

create relationship

review changes

publish dataset

77. PUBLIC VS EDITORIAL DATA

Conceptually:

Editorial Database

↓

Validation

↓

Published Knowledge Base

↓

Public Explorer

78. V1 EDITORIAL WORKFLOW

A full CMS is not required for V1, but the data structures should be importable/editable by the project owner.

79. V1 / V2 / V3 AUDIT

Strict prioritization is necessary because the project can easily become enormous.

80. V1 — MUST HAVE

Core experience:

responsive application shell

Timeline

Characters

Events

Locations

Basic Relationships

Map

War Explorer

Search

Focus

Source/Evidence foundation

Settings foundation

81. V1 TIMELINE

Must have:

zoom/detail levels

chronological navigation
event cards
event detail
contextual navigation
responsive behavior
loading/empty states

82. V1 CHARACTER SYSTEM

Must have:
character profile
relationships
timeline presence
event connections
source information
search
focus

83. V1 EVENT SYSTEM

Must have:
event detail
chronological position
participants
locations
related events
sources
timeline connection

84. V1 MAP

Must have:
locations
basic markers
location detail
event connection
responsive interaction

Basic journey path may be included if the dataset supports it cleanly.

85. V1 WAR

Must have:
war overview
day navigation
day events
participants
event links
timeline connection
map connection
source connection

Formation visualization is conditional on data readiness.

86. V1 SEARCH

Must have:

- global search
- entity type recognition
- relevant results
- deep links
- empty states
- mobile search

87. V1 EVIDENCE

Must have:

- Source entity
- source metadata
- evidence relationship
- contextual source drawer
- source detail
- claim-level provenance where needed
- uncertainty/status architecture

88. V1 SETTINGS

Must have:

- theme
- text size
- density
- motion
- visual character
- responsive behavior
- preference persistence

Some advanced settings can remain architectural placeholders.

89. V1 DESIGN

Must have:

- central design tokens
- responsive component system
- typography system
- semantic colors
- reusable cards
- reusable drawers/sheets
- loading states
- empty states
- error states
- accessibility foundation

90. V2

Potential V2:

- Progressive chronology / spoiler mode
- Advanced Formation Explorer
- Advanced relationship visualization

Character journey playback
Source comparison
Variant account comparison
Advanced research search
More sophisticated map layers
Richer analytics
Saved explorations
Bookmarks
Notes
Multilingual content

91. V3 / LONG-TERM

Potentially:
Research Mode
AI grounded Q&A
Source passage exploration
Advanced textual comparison
Contributor/editorial workflows
Dataset version history
Collaborative research
Advanced visualization
Public API

92. CRITICAL V1 EXCLUSION: AI Q&A;

Do not build AI Q&A into V1.

First:
structured knowledge
then:
evidence
then:
grounded AI

93. CRITICAL V1 EXCLUSION: 3D WORLD

Do not build a giant 3D Mahabharata world.

The product is a knowledge explorer, not an open-world game.

94. CRITICAL V1 EXCLUSION: BATTLE SIMULATION

Do not build animated battle simulations that imply speculative battlefield movement.

95. CRITICAL V1 EXCLUSION: STATIC ENCYCLOPEDIA

Do not turn the product into a giant encyclopedia page system.

The core value is connected exploration.

96. THE REAL PRODUCT DIFFERENTIATOR

The Mahabharata is represented as a connected, explorable knowledge graph across time, people, places, events, relationships, and sources.

The UI is the mechanism through which users explore that graph.

97. THE SIX PRIMARY LENSES

TIME

People / Events / War

SPACE

Locations / Map / Journeys

PEOPLE

Characters / Groups

RELATIONSHIPS

Family / Social / Conflict

EVENTS

Chronology / War / Consequences

EVIDENCE

Sources / Claims / Variants

Search and Focus operate across all six.

98. HOMEPAGE PHILOSOPHY

The experience should communicate:

Explore the Mahabharata through time, people, places, events and relationships.

99. NO DEAD ENDS RULE

Whenever the user encounters an entity, meaningful paths to related information should be available.

Character → Events / Timeline / Relationships / Family / War / Sources

Event → People / Location / Timeline / War / Sources

Location → Events / People / Map / Sources

Source → Claims / Entities / References

100. NO FALSE CONNECTIONS RULE

Only show relationships explicitly represented by the data.

Proximity on a timeline or map does not automatically imply a relationship.

101. NO FALSE PRECISION RULE

Do not invent:

exact dates

exact coordinates

exact sequence

troop positions

numerical statistics

causal relationships

when the data doesn't establish them.

102. NO FALSE AUTHORITY RULE

Do not make:

- an illustration look like a primary-source diagram
- an editorial interpretation look like a quotation
- an AI summary look like a source
- a disputed account look universally settled

103. NO VISUAL OVERLOAD RULE

Complexity should exist in the data architecture, not necessarily in the first screen.

Progressive disclosure is essential.

104. FIRST-TIME USER EXPERIENCE

A first-time visitor should quickly understand:

What is this? A Mahabharata explorer.

What can I do? Explore people, events, places and chronology.

How do I go deeper? Click anything.

Can I verify information? Yes, sources are attached throughout.

105. PRODUCT UX LOOP

DISCOVER



EXPLORE



ZOOM



CONNECT



VERIFY



RETURN



DISCOVER SOMETHING ELSE

106. EXAMPLE EXPLORATION LOOP

Search "Karna"



Character Profile



War Journey



Day 17



Event



Location



Map



Source



Related Character



Character Profile

107. AUDIT CONCLUSION

The product has a clear identity, coherent information architecture, identifiable shared systems, strong evidence architecture, responsive philosophy, and a consistent visual identity.

The main areas requiring formalization are:

data architecture

V1 scope

source ingestion workflow

108. BIGGEST RISK

The biggest risk is feature proliferation.

Almost every idea is genuinely cool, which makes the project particularly vulnerable to expanding beyond its core objective.

109. FEATURE DISCIPLINE

Use:

Amazing idea → V2

when an idea does not materially strengthen the first coherent product.

110. THE PRODUCT WE SHOULD BUILD FIRST

A beautiful, responsive, deeply interconnected Mahabharata knowledge explorer where users can navigate through time, people, events, places, relationships and the Kurukshetra War—and inspect the sources behind the information.

111. MASTER ARCHITECTURE

ONE KNOWLEDGE GRAPH



SHARED DATA MODELS



SHARED COMPONENT SYSTEM



MULTIPLE EXPLORATION LENSES



GLOBAL FOCUS + SEARCH



CROSS-CONTEXT NAVIGATION



EVIDENCE / SOURCES



PERSONALIZED PRESENTATION

112. FIVE FOUNDATIONAL LAYERS

Data → Evidence → Exploration → Presentation → User State

113. ARCHITECTURAL LOCK

One Knowledge Graph
Shared Data Models
Shared Component System
Multiple Exploration Lenses
Global Focus + Search
Cross-Context Navigation
Evidence / Sources
Personalized Presentation

114. DATA MODEL — MASTER VERSION

Core entities:

Character
Event
Location
Group
Relationship
FamilyRelationship
War
WarDay
Formation
Source
Claim
Evidence

Potential future:

Journey
DatasetVersion
EditorialNote
VariantAccount

115. RELATIONSHIP MODEL

Relationships should be first-class.

Examples:

Character → Character
Character → Event
Character → Location
Character → Group
Event → Event
Event → Location
Event → WarDay
Formation → Event
Claim → Entity
Evidence → Claim
Evidence → Source

116. NO HARD-CODED RELATIONSHIPS

Relationships should be represented as data wherever possible, not hardcoded into page logic.

117. DATA-DRIVEN UI

The frontend should render based on available structured data.

If formation is null, do not render an empty Formation section.

If sources are empty, do not render a broken source drawer.

118. FUTURE-PROOFING

If new records are added, the application should not need redesign.

The data grows; the visualization system remains stable.

119. SOURCE EXPANSION FIT

The architecture supports:

original PDF dataset

+

Source A

+

Source B

+

Source C

+

future credible sources

→

one knowledge base

120. RESEARCH WORKFLOW

Question

↓

Source Discovery

↓

Source Quality Check

↓

Extraction

↓

Cross-check

↓

Structuring

↓

Claim

↓

Evidence

↓

Database

121. RESEARCH WORKFLOW OUTSIDE BASE44

Initially, research and source verification should remain a deliberate workflow between the project owner and research process.

Base44's job is to render and manage the structured knowledge system, not autonomously research the Mahabharata.

122. DATA QUALITY GATE

Before information enters the published dataset:

Source identified?

↓

Credible?

↓

Specific reference?

↓

Claim extracted?

↓

Conflicting accounts checked?

↓

Structured?

↓

Published

123. MASTER DESIGN PRINCIPLES

01 — Modern first, traditional second.

02 — Information is the hero.

03 — The interface should feel Indian without becoming stereotypically "ancient."

04 — No excessive warmth, parchment, brown, gold, or ornamental clutter.

05 — Responsive is fundamental, not a desktop adaptation.

06 — One knowledge graph, many exploration lenses.

07 — Reuse systems instead of duplicating them.

08 — Data before decoration.

09 — Evidence before certainty.

10 — Never fabricate missing information.

11 — Preserve uncertainty and source differences.

12 — Progressive disclosure keeps complexity manageable.

13 — Accessibility is foundational.

14 — The user controls presentation, not facts.

15 — Build architecture first, thematic assets second.

124. THE REAL PRODUCT DIFFERENTIATOR — REFINED

The product is not primarily the artwork, not primarily the timeline, and not primarily the family tree.

Its differentiator is:

ONE WORLD

↓

MANY LENSES

↓

CONNECTED EXPLORATION

125. USER EXPERIENCE AS A CONNECTED GRAPH

The application should feel like a world in which any meaningful object can lead to another meaningful object without dead ends.

126. EVIDENCE AS A CROSS-CUTTING SYSTEM

Evidence should be available in context rather than isolated on a bibliography page.

127. SETTINGS AS PRESENTATION CONTROL

Settings customize the experience, not the truth.

They can change:

appearance

density

text size

motion

visual character

chronological reveal

They cannot change:

source provenance

factual relationships

event ordering

geographic evidence

uncertainty

canonical data

128. ASSET STRATEGY

Build generic components first.

Then progressively replace selected visual slots with Mahabharata-inspired assets.

129. RESPONSIVE-FIRST STRATEGY

Every major experience must be designed for desktop and mobile as first-class layouts, including landscape and portrait orientations.

130. SOURCE STRATEGY

The original PDF is the seed dataset.

It is not assumed to be complete.

Future expansion must use credible, identifiable sources and preserve provenance.

131. V1 STRATEGY

Build a deliberately selected representative subset of the data first, containing enough complexity to exercise:

Timeline

Character

Event

Relationship

Map

War

Evidence

Search

Then expand the knowledge base.

132. WHY REPRESENTATIVE DATA FIRST

If a large dataset exposes a flawed relationship model, fixing it after thousands of records have been loaded is painful.

Instead:

Small representative dataset

↓

Build

↓

Test

↓

Architecture correction

↓

Expand dataset

133. FINAL AUDIT VERDICT

The project has moved far beyond the original idea of "make an interactive mind map."

It is now best understood as:

A beautiful, responsive, source-aware, interconnected Mahabharata knowledge explorer.

The original PDF remains the seed.

The product architecture is now:

Data → Evidence → Exploration → Presentation → User State.

The next step should be a formal technical consolidation of the knowledge graph, entity model, relationship model, claims/evidence model, and information architecture before translating the work into Base44 implementation instructions.